



# MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL(M.P.)

DEPARTMENT OF MATHEMATICS BIOINFORMATICS & COMPUTER APPLICATIONS

## SUPERMARKET INVENTORY MANAGEMENT SYSTEM

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# WHY?

## WHY DID WE CHOOSE THIS SPECIFIC PROJECT?

Manual inventory in supermarkets leads to huge losses due to expiry (perishables) and theft. This project solves a real-world problem of financial leakage and operational inefficiency

## WHY IS AN AUTOMATED SYSTEM BETTER THAN EXCEL/MANUAL RECORDS FOR A SUPERMARKET?

Excel doesn't give real-time alerts for low stock or expiry. It doesn't support concurrent users (multiple cashiers), and it lacks role-based security (admin vs. staff)

## WHY A SUPERMARKET?

Supermarkets present a unique challenge called "Perishability"(goods that have a limited shelf life and will spoil). Unlike a hardware store or bookstore, a supermarket deals with items that expire . This forces the system to handle FEFO (First Expired, First Out) logic, which adds significant complexity to the project

# PROJECT SCOPE

## FUNCTIONAL SCOPE (THE "WHAT")

- **Inventory & Stock Management:** Comprehensive tracking of products across 31 tables, including batch-wise management and automated stock reconciliation.
- **Customer Relationship Management (CRM):** A tiered loyalty program that tracks purchase history and manages reward points automatically.

## ANALYTICAL SCOPE (THE "INSIGHTS")

- **Predictive Logistics:** Using the "Festival Demand" logic to identify seasonal trends and optimize stock levels before peak periods.

## TECHNICAL / ARCHITECTURE SCOPE (THE "HOW")

- **Schema Normalization:** Designing a 3rd Normal Form (3NF) architecture to ensure data integrity and eliminate redundancy across 31 tables.
- **Automated Data Integrity:** Utilizing Triggers to enforce complex business rules (like preventing sales of expired items) that simple constraints cannot handle.
- **Security & Data Abstraction:** Using Views to provide a "Need-to-Know" access model, ensuring sensitive data (like profit margins) is hidden from standard users.

## OPERATIONAL SCOPE (THE "CONSTRAINTS")

- **Risk Mitigation:** Automated Low-Stock Alerts and system-level blocks on Expired Inventory via the trigger layer.
- **Logistics Strategy:** Implementation of the FEFO (First-Expired, First-Out) methodology to minimize inventory wastage.

# OBJECTIVES OF THE PROJECT

## AUTOMATION & EFFICIENCY

- **Make operations easier:** Automate billing and restocking processes to save time and reduce manual work.
- **Synchronize Data:** Keep all data updated instantly across all 31 tables using triggers, so that stock information is always correct and there are no mismatches.

## STRATEGIC INTELLIGENCE

- **Provide Actionable Insights:** To generate specialized management reports via Views, focusing on profitability, staff performance, and inventory health.
- **Forecasting & Planning:** To analyze historical sales data (Festival Demand) to help management prepare for peak shopping seasons.

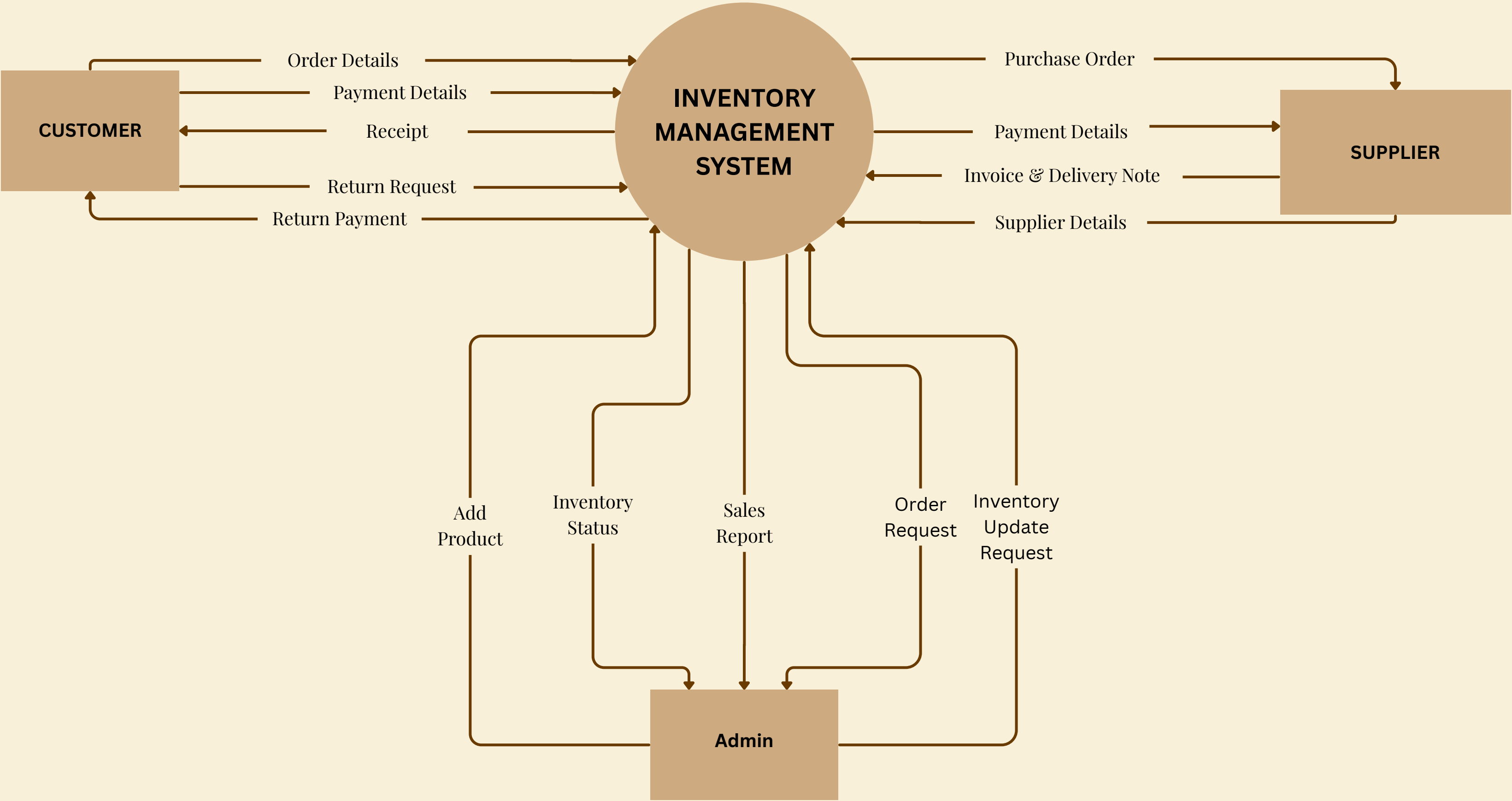
## RELIABILITY & INTEGRITY

- **Enforce Data Standards:** Keep the database in 3NF form to avoid Redundant data and maintain proper structure.
- **System-Level Validation:** Use strict triggers to stop errors like selling expired products.

## CUSTOMER & RESOURCE MANAGEMENT

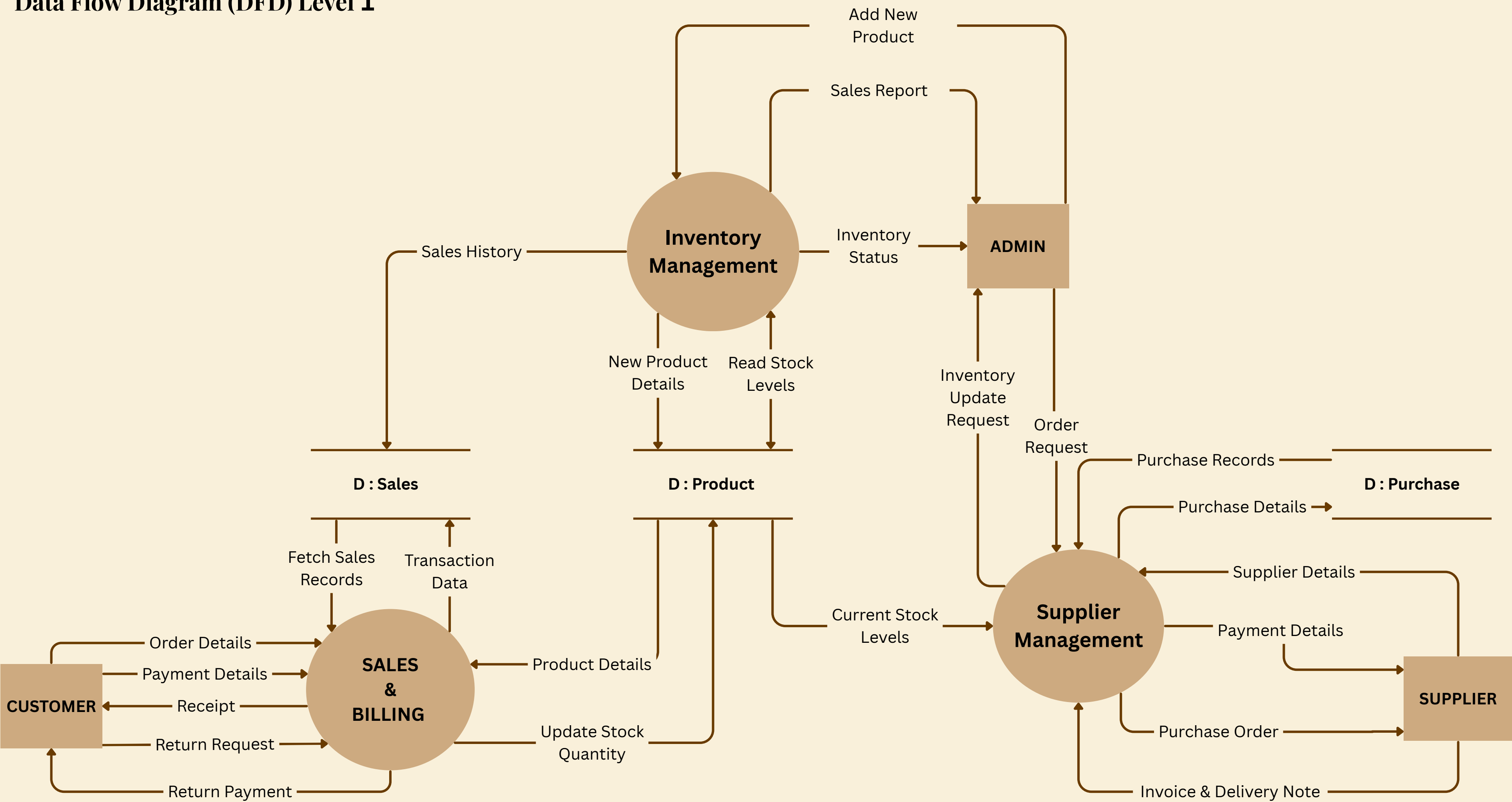
- **Personalize Experience:** Create a smart loyalty system that automatically tracks customer activity and gives rewards without manual work.
- **Optimize Logistics:** Reduce waste and old stock by using the FEFO (First-Expired, First-Out) method at the database level.

**Data Flow Diagram (DFD) Level O**





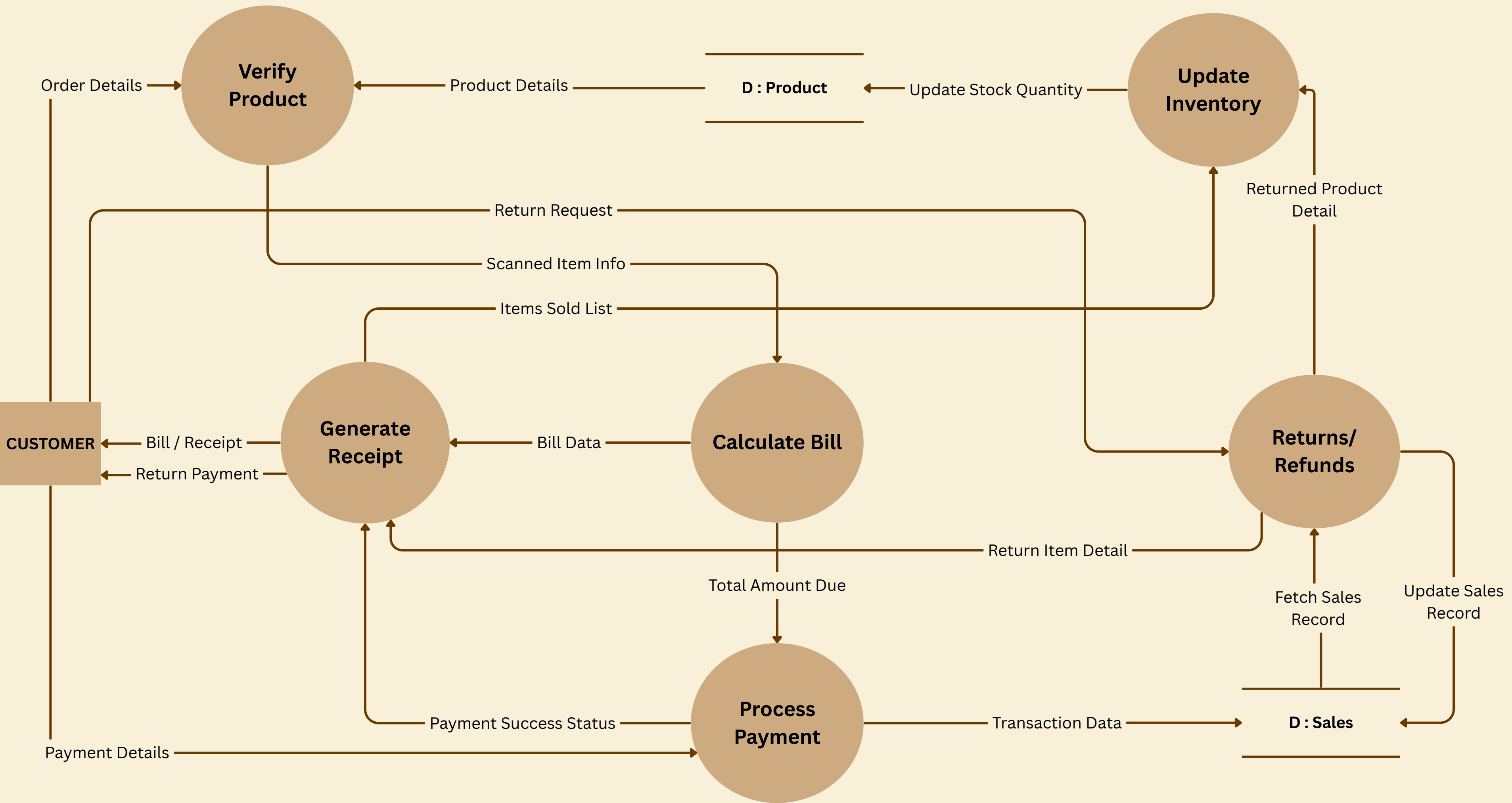
**Data Flow Diagram (DFD) Level 1**



## **Data Flow Diagram(DFD) Level 2**

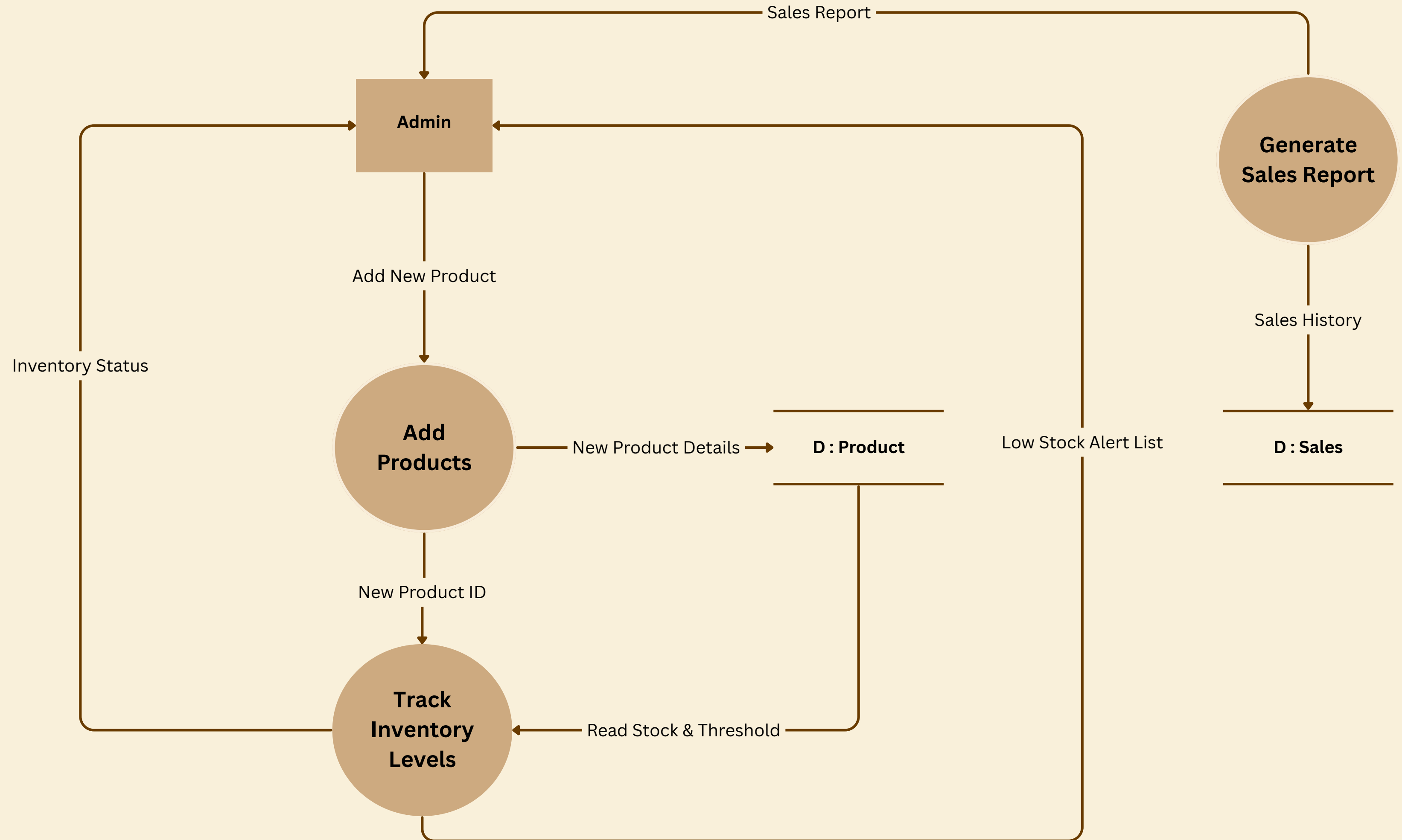
1. Sales & Billing Process
2. Inventory Management Process
3. Supplier Management Process

# Level 2 DFD for "Sales & Billing"

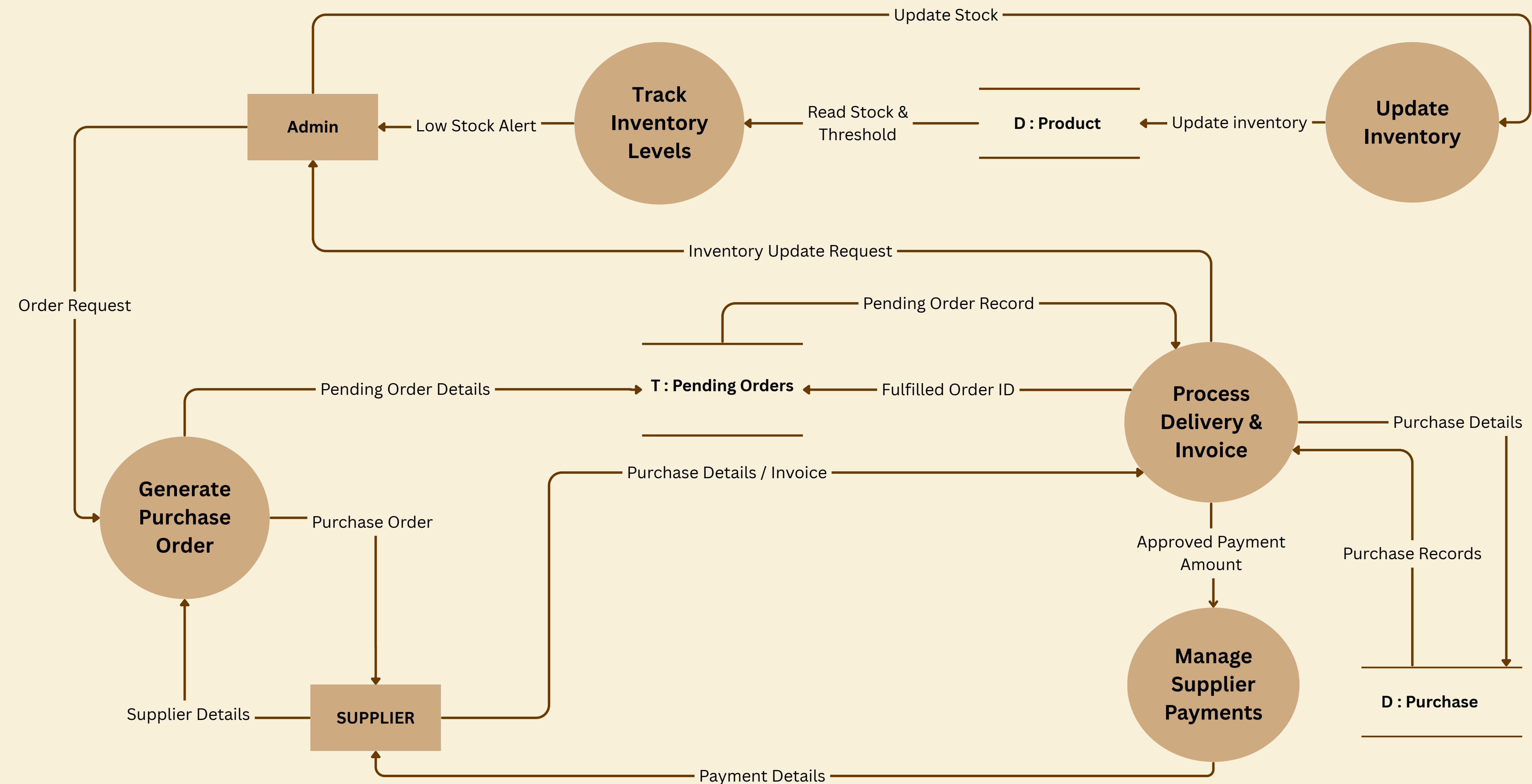




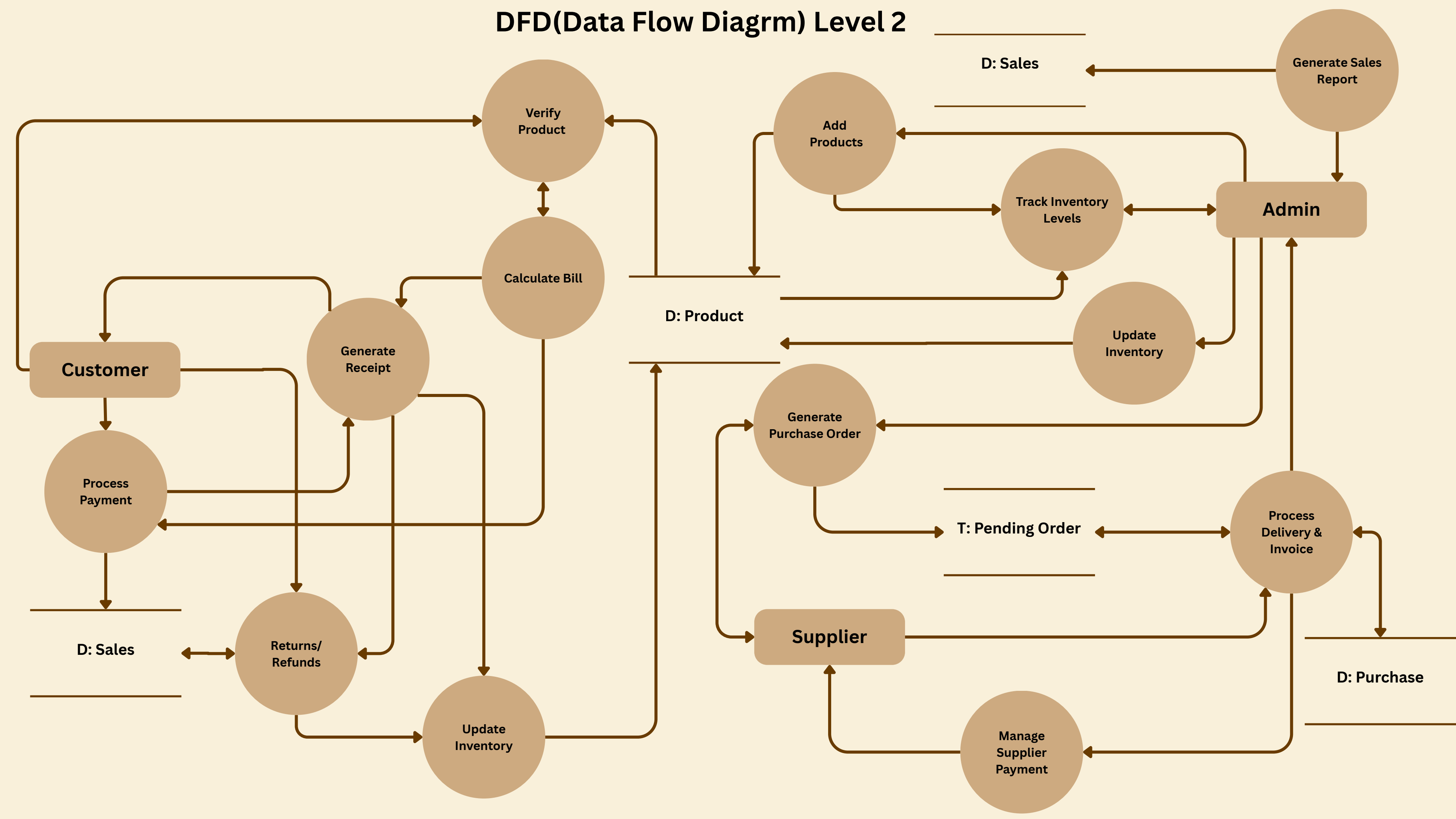
# Level 2 DFD for "Inventory Management"



# Level 2 DFD for "Supplier Management"



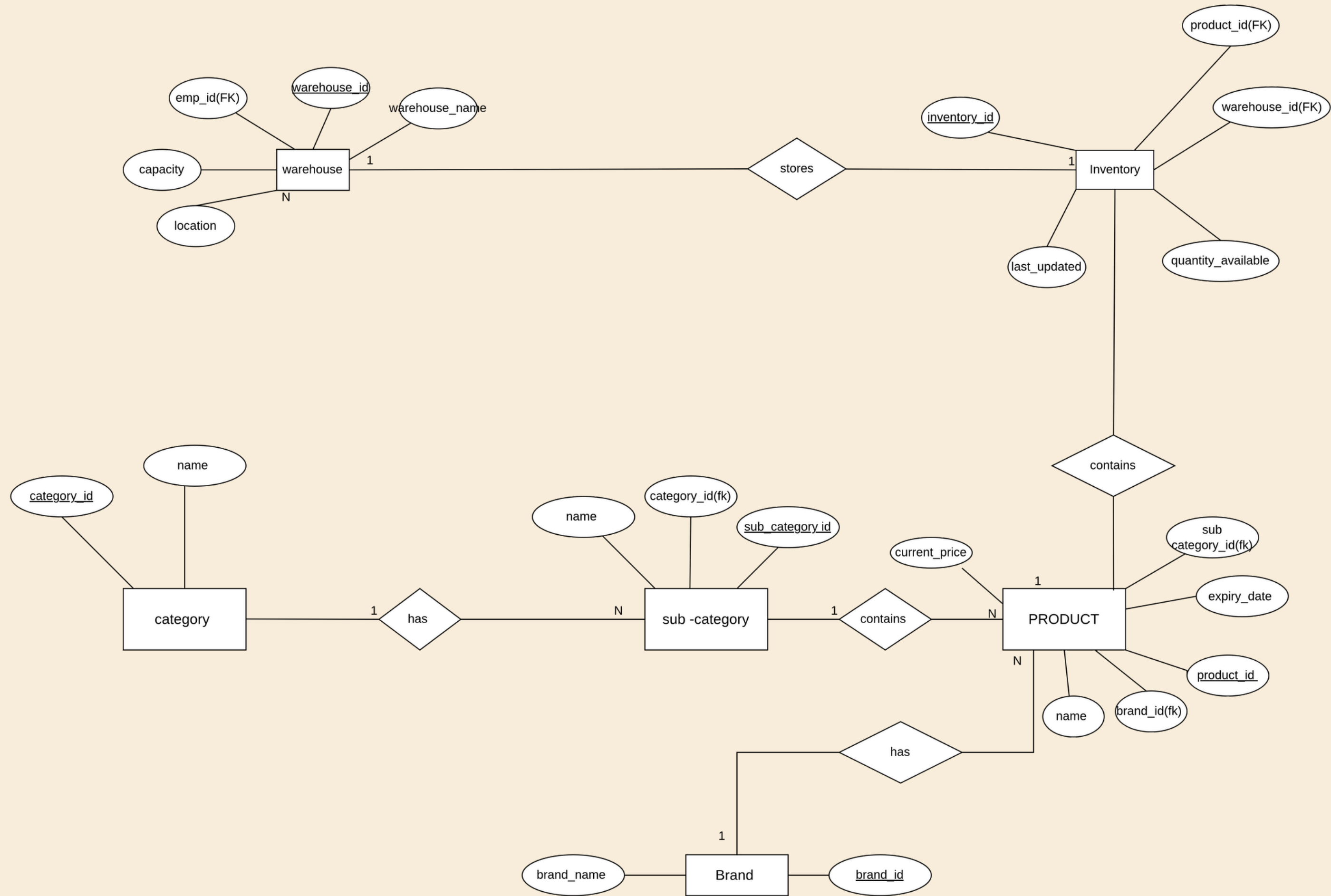
# DFD(Data Flow Diagram) Level 2



# ENTITY RELATIONSHIP MODULES

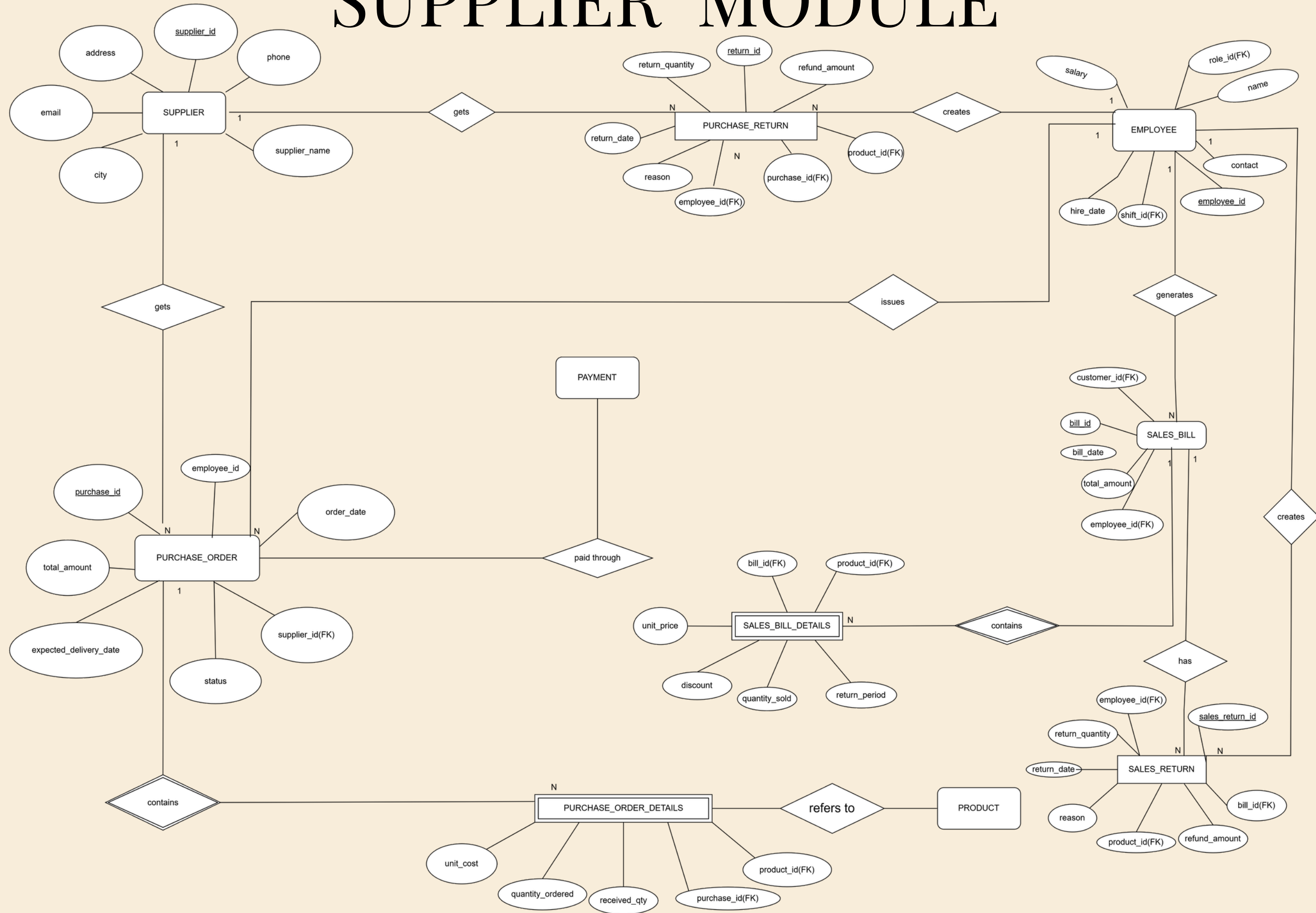
1. Product Module
2. Supplier Module
3. Sales Module
4. Employee Module
5. Marketing & Profit Module

# PRODUCT MODULE

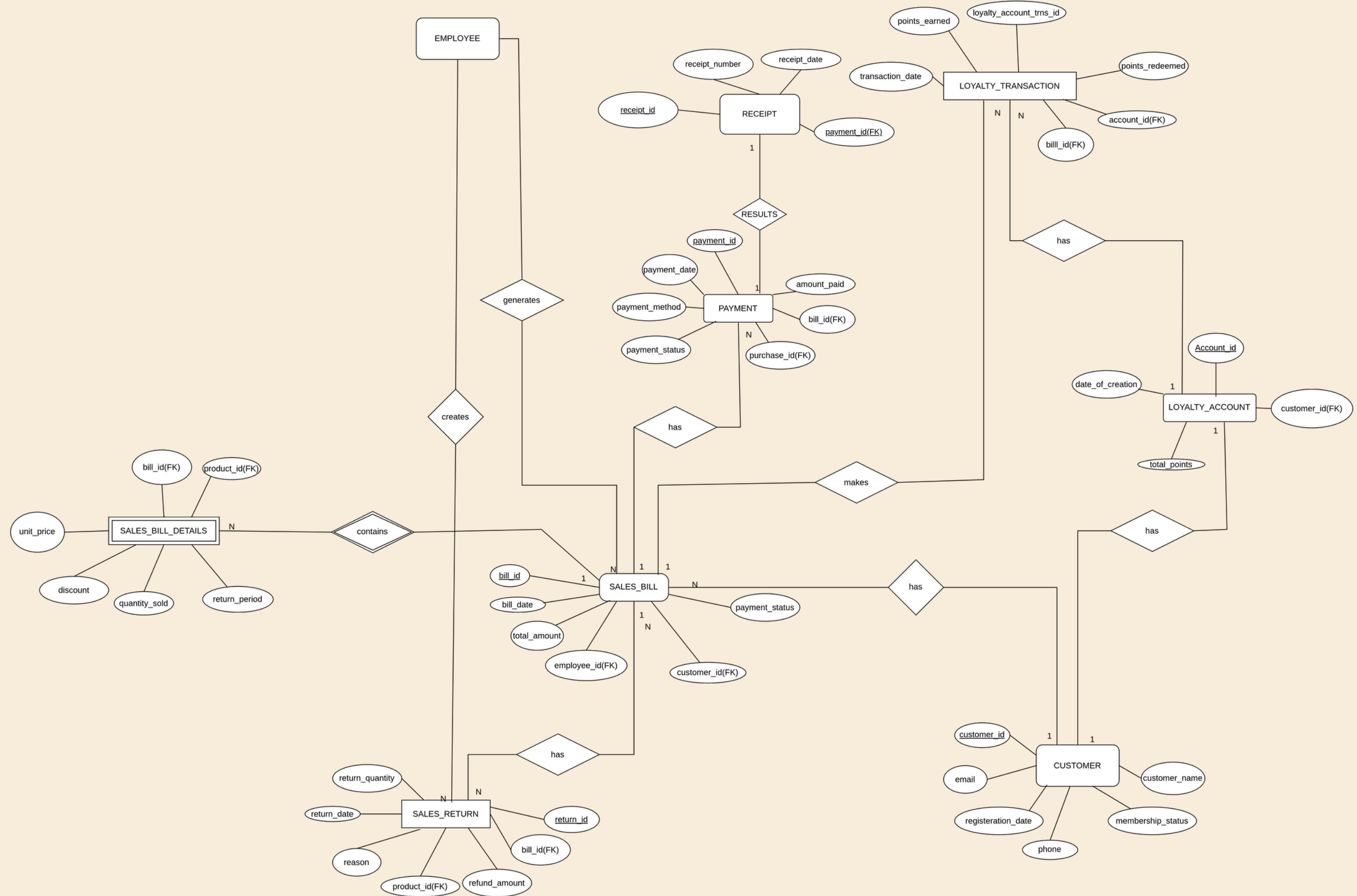




# SUPPLIER MODULE

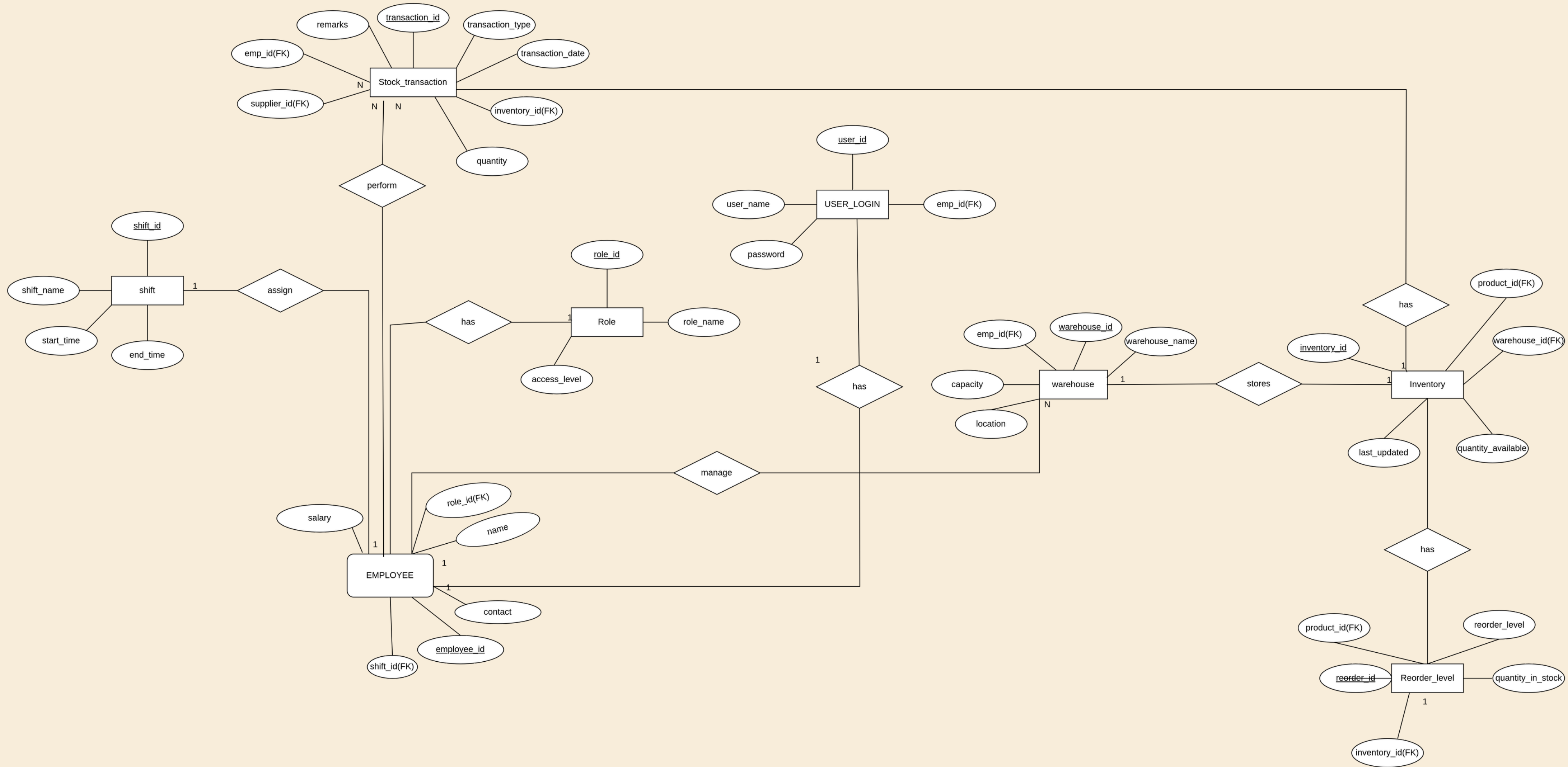


# SALES MODULE



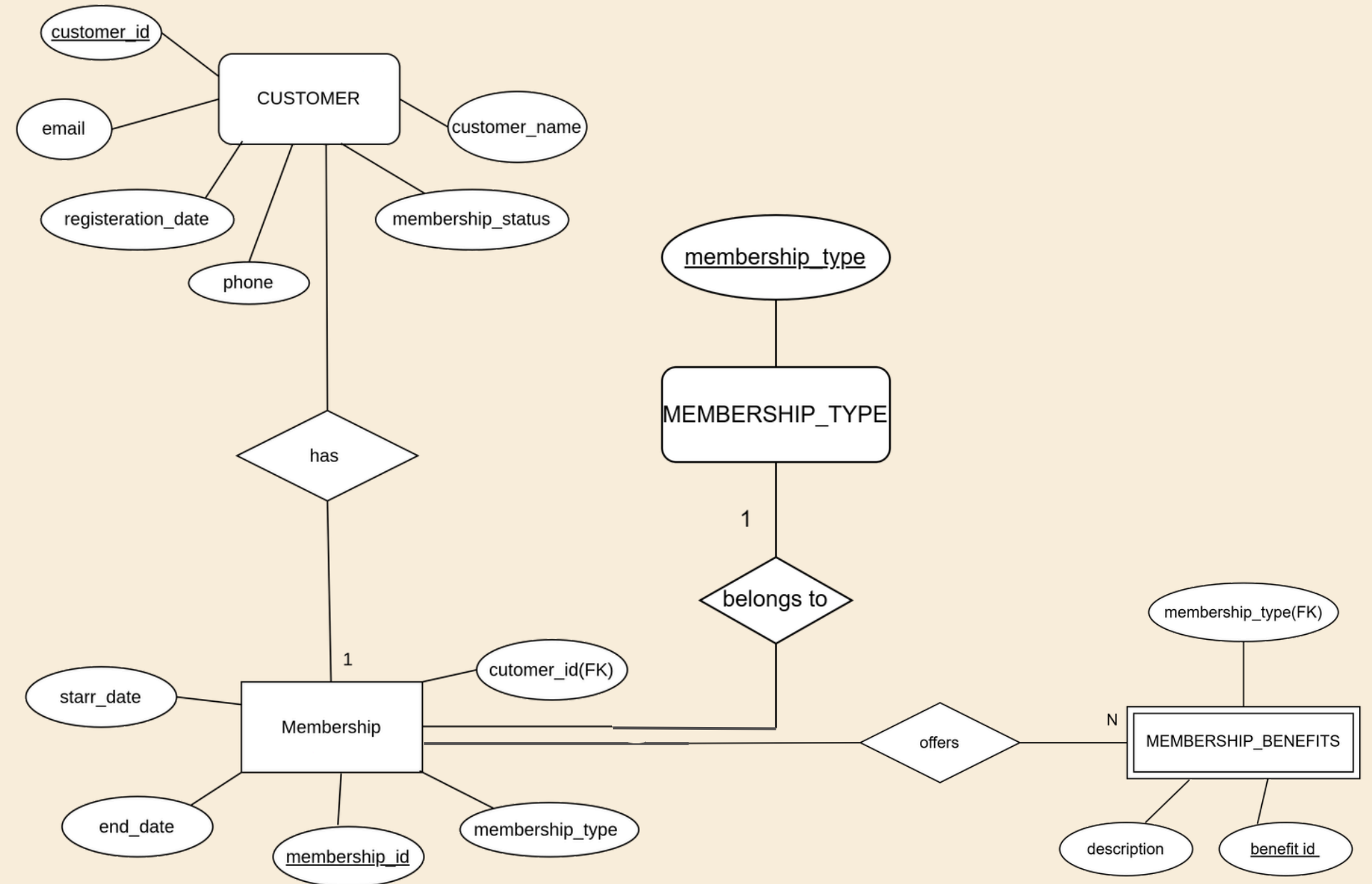
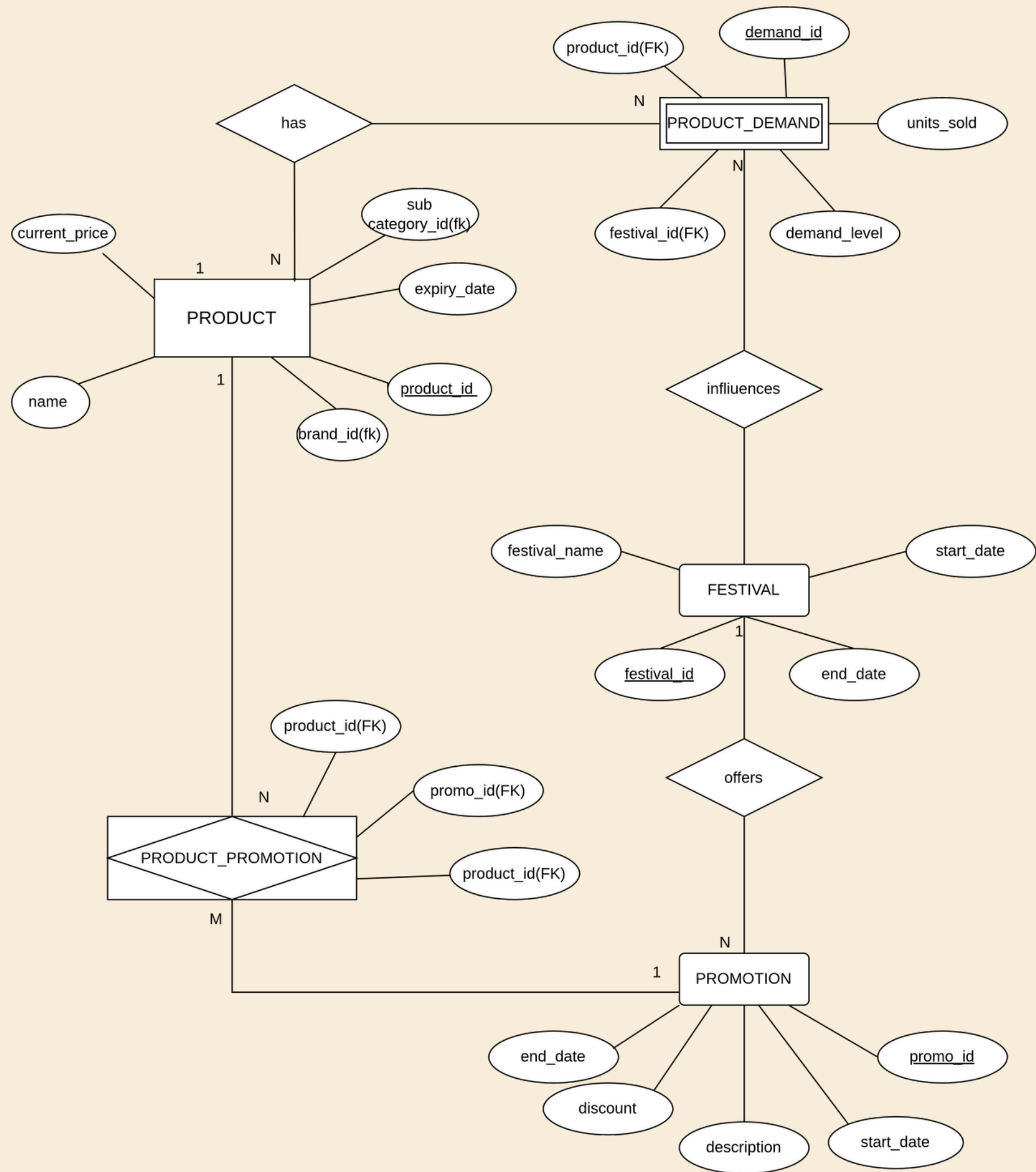


# EMPLOYEE MODULE

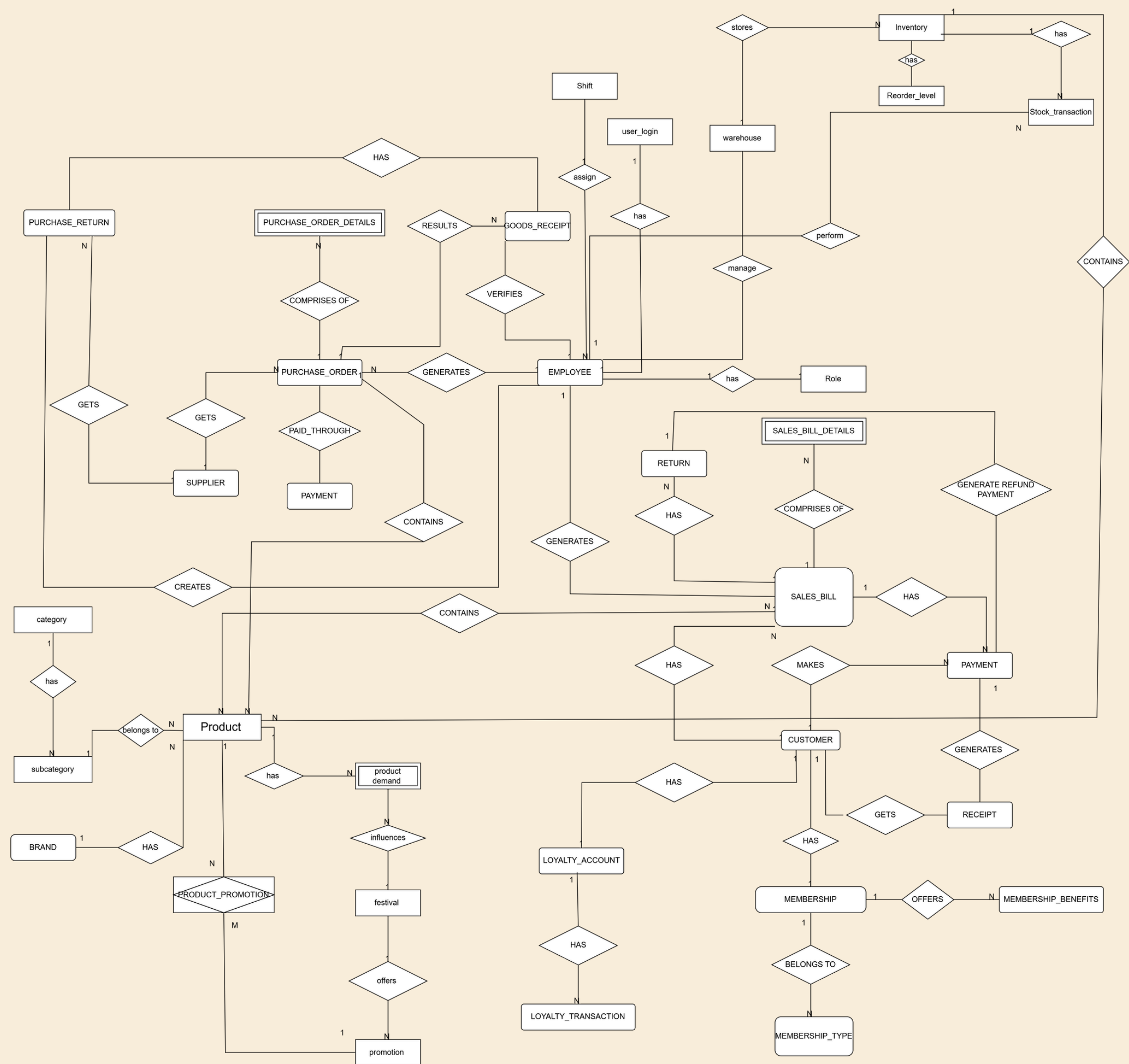




# MARKETING AND PROFIT MODULE



# Entity-Relationship(ER) Diagram



# DATABASE NORMALIZATION

1NF: REMOVED REPEATING GROUPS ➤ ATOMIC ATTRIBUTES

2NF: ELIMINATED PARTIAL DEPENDENCY ➤ SEPARATED DEPENDENT DATA

3NF: REMOVED TRANSITIVE DEPENDENCY ➤ CREATED INDEPENDENT TABLES

## PROJECT IMPLEMENTATION

MEMBERSHIP DETAILS SEPARATED INTO MEMBERSHIP\_TYPE TABLE

SALES SPLIT INTO SALES\_BILL AND SALES\_BILL\_DETAILS

PROPER PRIMARY KEYS & FOREIGN KEYS USED FOR RELATIONSHIPS

## **OUTCOME:**

NO DATA REDUNDANCY

IMPROVED DATA INTEGRITY

EASIER UPDATES AND MAINTENANCE

# DATA DICTIONARY

Supermarket Inventory Management System

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| MODULES                   | TABLES                                                                                                        | NUMBER OF TABLES | PURPOSE                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRODUCT MODULE            | Product, brand, category, sub-category, inventory, warehouse                                                  | 6                | Manages product information, classification, and stock. organizes products into categories and tracks their availability in warehouses through inventory. |
| SUPPLIER MODULE           | Supplier, purchase order,purchase_order _detail, purchase_return                                              | 4                | Handles supplier details and purchasing process, including order placement, item tracking, and return of defective goods.                                 |
| SALES & BILLING MODULE    | Sales_bill, sales_bill_detail, sales_return, receipt, payment, Customer, loyalty_account, loyalty_transaction | 8                | Manages sales transactions, billing, payments, and returns. Also maintains customer data and loyalty points system.                                       |
| EMPLOYEE MODULE           | Employee, role, shift, user- login, stock_transaction, reorder_level                                          | 6                | Manages employee data, roles, shifts, and system access. Also tracks stock movement and maintains reorder levels.                                         |
| PROFIT & MARKETING MODULE | Product-demand, promotion, product-promotion, membership, membership benfits, membership_type, festival       | 7                | Supports marketing and analysis through promotions, demand tracking, and membership benefits to improve customer engagement.                              |



1. PRODUCT

| Column Name     | Data Type | Key | Constraints | Description             |
|-----------------|-----------|-----|-------------|-------------------------|
| product_id      | INT       | PK  |             | Unique product          |
| name            | VARCHAR   |     | NOT NULL    | Product name            |
| current_price   | DECIMAL   |     | CHECK > 0   | Current selling price   |
| sub_category_id | INT       | FK  | NOT NULL    | References SUB_CATEGORY |
| brand_id        | INT       | FK  |             | References BRAND        |

2. CATEGORY

| Column Name | Data Type | Key | Constraints | Description        |
|-------------|-----------|-----|-------------|--------------------|
| category_id | INT       | PK  |             | Unique category ID |
| name        | VARCHAR   | UK  | NOT NULL    | Category name      |

3. SUB\_CATEGORY

| Column Name     | DataType | Key | Constraints | Description         |
|-----------------|----------|-----|-------------|---------------------|
| sub_category_id | INT      | PK  |             | Unique subcategory  |
| name            | VARCHAR  |     | NOT NULL    | Subcategory name    |
| category_id     | INT      | FK  | NOT NULL    | References CATEGORY |

4. BRAND

| Column Name | Data Type | Key | Constraints      | Description |
|-------------|-----------|-----|------------------|-------------|
| brand_id    | INT       | PK  |                  | Brand ID    |
| brand_name  | VARCHAR   | UK  | UNIQUE, NOT NULL | Brand name  |

5. INVENTORY

| Column Name        | Data Type | Key | Constraints          | Description         |
|--------------------|-----------|-----|----------------------|---------------------|
| inventory_id       | SERIAL    | PK  |                      | Unique inventory ID |
| product_id         | INT       | FK  | NOT NULL             | Product reference   |
| warehouse_id       | INT       | FK  | NOT NULL             | Warehouse reference |
| batch_number       | VARCHAR   |     | NOT NULL             | Batch identifier    |
| mfg_date           | DATE      |     |                      | Manufacturing date  |
| expiry_date        | DATE      |     | CHECK valid          | Expiry date         |
| quantity_available | INT       |     | ≥ 0                  | Stock quantity      |
| last_updated       | DATE      |     | DEFAULT CURRENT_DATE | Last update         |

## 6. WAREHOUSE

| Column Name    | Data Type | Key | Constraints      | Description  |
|----------------|-----------|-----|------------------|--------------|
| warehouse_id   | SERIAL    | PK  |                  | Warehouse ID |
| warehouse_name | VARCHAR   | UK  | UNIQUE, NOT NULL | Name         |
| location       | VARCHAR   |     | NOT NULL         | Location     |
| capacity       | INT       |     | > 0              | Capacity     |
| employee_id    | INT       | FK  |                  | Manager      |

## 7. STOCK\_TRANSACTION

| Column Name      | Data Type | Key | Constraints     | Description          |
|------------------|-----------|-----|-----------------|----------------------|
| transaction_id   | SERIAL    | PK  |                 | Transaction ID       |
| transaction_type | VARCHAR   |     | IN/OUT/TRANSFER | Type of movement     |
| quantity         | INT       |     | > 0 DEFAULT NOW | Quantity moved       |
| transaction_date | TIMESTAMP |     | NOT NULL NOT    | Date of transaction  |
| inventory_id     | INT       | FK  | NULL            | Inventory reference  |
| employee_id      | INT       | FK  |                 | Employee responsible |
| supplier_id      | INT       | FK  |                 | Supplier reference   |
| remarks          | TEXT      |     |                 | Additional notes     |

## 8. REORDER\_LEVEL

| Column Name     | Data Type | Key | Constraints | Description |
|-----------------|-----------|-----|-------------|-------------|
| reorder_id      | SERIAL    | PK  |             | ID          |
| product_id      | INT       | FK  | NOT NULL    | Product     |
| warehouse_id    | INT       | FK  | NOT NULL    | Warehouse   |
| min_quantity    | INT       |     | > 0 > min   | Minimum     |
| target_quantity | INT       |     |             | Target      |

## 9. SUPPLIER

| Column Name   | Data Type | Key | Constraints       | Description |
|---------------|-----------|-----|-------------------|-------------|
| supplier_id   | INT       | PK  |                   | Supplier ID |
| supplier_name | VARCHAR   |     | NOT NULL          | Name        |
| phone         | VARCHAR   | UK  | UNIQUE, 10 digits | Contact     |
| email         | VARCHAR   | UK  | UNIQUE            | Email       |
| address       | VARCHAR   |     | NOT NULL NOT      | Address     |
| city          | VARCHAR   |     | NULL              | City        |



10. PURCHASE\_ORDER

| Column Name            | Data Type | Key | Constraints | Description       |
|------------------------|-----------|-----|-------------|-------------------|
| purchase_id            | INT       | PK  |             | Order ID          |
| order_date             | DATE      |     | NOT NULL ≥  | Order date        |
| total_amount           | DECIMAL   |     | 0 ≥         | Total cost        |
| expected_delivery_date | DATE      |     | order_date  | Expected delivery |
| status                 | VARCHAR   |     | ENUM NOT    | Order status      |
| supplier_id            | INT       | FK  | NULL NOT    | Supplier          |
| employee_id            | INT       | FK  | NULL        | Managed by        |

11. PURCHASE\_ORDER\_DETAILS

| Column Name       | Data Type | Key   | Constraints | Description   |
|-------------------|-----------|-------|-------------|---------------|
| purchase_id       | INT       | PK/FK |             | Order ID      |
| product_id        | INT       | PK/FK |             | Product       |
| quantity_ordered  | INT       |       | > 0         | Ordered qty   |
| received_quantity | INT       |       | ≥ 0         | Received qty  |
| unit_cost         | DECIMAL   |       | > 0         | Cost per unit |

12. PURCHASE\_RETURN

| Column Name        | Data Type | Key | Constraints          | Description |
|--------------------|-----------|-----|----------------------|-------------|
| purchase_return_id | INT       | PK  |                      | Return ID   |
| return_date        | DATE      |     | DEFAULT CURRENT_DATE | Date        |
| return_quantity    | INT       |     | > 0 ≥ 0              | Quantity    |
| reason             | TEXT      |     |                      | Reason      |
| refund_amount      | DECIMAL   |     |                      | Refund      |
| product_id         | INT       | FK  | NOT NULL             | Product     |
| purchase_id        | INT       | FK  | NOT NULL             | Order       |
| employee_id        | INT       | FK  | NOT NULL             | Employee    |



13. SALES\_BILL

| Column Name    | Data Type | Key | Constraints | Description      |
|----------------|-----------|-----|-------------|------------------|
| bill_id        | INT       | PK  |             | Unique bill      |
| bill_date      | TIMESTAMP |     | DEFAULT NOW | Bill timestamp   |
| total_amount   | DECIMAL   |     |             | Total bill value |
| employee_id    | INT       | FK  | NOT NULL    | Salesperson      |
| customer_id    | INT       | FK  |             | Customer         |
| payment_status | VARCHAR   |     | ENUM values | Payment status   |

14. SALES\_BILL\_DETAILS

| Column Name   | Data Type | Key   | Constraints | Description      |
|---------------|-----------|-------|-------------|------------------|
| bill_id       | INT       | PK/FK |             | Bill reference   |
| product_id    | INT       | PK/FK |             | Product sold     |
| quantity_sold | INT       |       | > 0         | Quantity sold    |
| unit_price    | DECIMAL   |       | > 0         | Price per unit   |
| discount      | DECIMAL   |       | ≥ 0         | Discount applied |

15. PAYMENT

| Column Name    | Data Type | Key | Constraints | Description      |
|----------------|-----------|-----|-------------|------------------|
| payment_id     | INT       | PK  |             | Payment ID       |
| payment_date   | TIMESTAMP |     | DEFAULT NOW | Payment time     |
| payment_method | VARCHAR   |     | ENUM ENUM > | Mode (UPI, Cash) |
| payment_status | VARCHAR   |     | 0           | Status           |
| amount_paid    | DECIMAL   |     |             | Amount paid      |
| bill_id        | INT       | FK  |             | Linked bill      |

16. RECEIPT

| Column Name    | Data Type | Key | Constraints         | Description |
|----------------|-----------|-----|---------------------|-------------|
| receipt_id     | SERIAL    | PK  |                     | Receipt ID  |
| receipt_number | VARCHAR   | UK  | UNIQUE DEFAULT      | Receipt No  |
| receipt_date   | DATE INT  |     | CURRENT_DATE UNIQUE | Date        |
| payment_id     | INT       | FK  | UNIQUE              | Payment     |
| bill_id        |           | FK  |                     | Bill        |

17. SALES\_RETURN

| Column Name     | Data Type | Key | Constraints             | Description |
|-----------------|-----------|-----|-------------------------|-------------|
| sales_return_id | INT       | PK  |                         | Return ID   |
| bill_id         | INT       | FK  | NOT NULL                | Bill        |
| product_id      | INT       | FK  | NOT NULL                | Product     |
| return_date     | DATE      |     | DEFAULT<br>CURRENT_DATE | Date        |
| return_quantity | INT       |     | > 0                     | Quantity    |
| refund_amount   | DECIMAL   |     | ≥ 0                     | Refund      |
| reason          | TEXT      |     |                         | Reason      |
| employee_id     | INT       | FK  |                         | Employee    |

18. CUSTOMER

| Column Name       | Data Type | Key | Constraints                  | Description                 |
|-------------------|-----------|-----|------------------------------|-----------------------------|
| customer_id       | INT       | PK  | NOT NULL                     | Unique ID for each customer |
| name              | VARCHAR   |     | NOT NULL                     | Customer's full name        |
| email             | VARCHAR   |     | CHECK (valid format)         | Customer email              |
| phone             | VARCHAR   | UK  | UNIQUE, CHECK<br>(10 digits) | Customer contact number     |
| registration_date | DATE      |     | DEFAULT<br>CURRENT_DATE      | Date of registration        |

19. LOYALTY\_ACCOUNT

| Column Name  | Data Type | Key | Constraints | Description     |
|--------------|-----------|-----|-------------|-----------------|
| account_id   | SERIAL    | PK  |             | Loyalty account |
| customer_id  | INT       | FK  | UNIQUE      | Linked customer |
| total_points | INT       |     | ≥ 0         | Points balance  |
| start_date   | DATE      |     | DEFAULT NOW | Start date      |
| end_date     | DATE      |     |             | Expiry          |

20. LOYALTY\_TRANSACTION

| Column Name            | Data Type | Key | Constraints     | Description     |
|------------------------|-----------|-----|-----------------|-----------------|
| loyalty_transaction_id | SERIAL    | PK  |                 | Transaction ID  |
| account_id             | INT       | FK  | NOT        NULL | Loyalty account |
| bill_id                | INT       | FK  |                 | Related bill    |
| transaction_date       | TIMESTAMP |     | DEFAULT NOW     | Date            |
| points_earned          | INT       |     | ≥ 0             | Points added    |
| points_redeemed        | INT       |     | ≥ 0             | Points used     |
| transaction_type       | VARCHAR   |     | ENUM            | Earn/Redeem     |
| remarks                | TEXT      |     |                 | Notes           |

21. MEMBERSHIP\_TYPE

| Column Name     | Data Type | Key | Constraints | Description |
|-----------------|-----------|-----|-------------|-------------|
| membership_type | VARCHAR   | PK  |             | Type name   |

22. MEMBERSHIP\_BENEFITS

| Column Name     | Data Type | Key | Constraints | Description     |
|-----------------|-----------|-----|-------------|-----------------|
| benefit_id      | INT       | PK  |             | Benefit ID      |
| membership_type | VARCHAR   | FK  | NOT NULL    | Type            |
| description     | TEXT      |     | NOT NULL    | Benefit details |

23. MEMBERSHIP

| Column Name     | Data Type | Key | Constraints | Description   |
|-----------------|-----------|-----|-------------|---------------|
| membership_id   | INT       | PK  |             | Membership ID |
| customer_id     | INT       | FK  | NOT NULL    | Customer      |
| membership_type | VARCHAR   | FK  | NOT NULL    | Type          |
| start_date      | DATE      |     | NOT NULL    | Start date    |

24. PROMOTION

| Column Name | Data Type | Key | Constraints | Description |
|-------------|-----------|-----|-------------|-------------|
| promo_id    | INT       | PK  |             | Promo ID    |
| promo_name  | VARCHAR   | UK  | UNIQUE      | Name        |
| discount    | DECIMAL   |     | > 0         | Discount    |
| start_date  | DATE      |     | NOT NULL    | Start       |
| end_date    | DATE      |     |             | End         |

25. PRODUCT\_PROMOTION

| Column Name | Data Type | Key   | Constraints | Description |
|-------------|-----------|-------|-------------|-------------|
| product_id  | INT       | PK/FK |             | Product     |
| promo_id    | INT       | PK/FK |             | Promotion   |

26. FESTIVAL

| Column Name   | Data Type | Key | Constraints | Description |
|---------------|-----------|-----|-------------|-------------|
| festival_id   | INT       | PK  |             | Festival ID |
| festival_name | VARCHAR   | UK  | UNIQUE      | Name        |
| start_date    | DATE      |     | NOT NULL    | Start       |
| end_date      | DATE      |     |             | End         |

27. PRODUCT\_DEMAND

| Column Name  | Data Type | Key | Constraints     | Description |
|--------------|-----------|-----|-----------------|-------------|
| demand_id    | INT       | PK  |                 | Demand ID   |
| product_id   | INT       | FK  | NOT NULL NOT    | Product     |
| festival_id  | INT       | FK  | NULL ≥ 0        | Festival    |
| units_sold   | INT       |     | Low/Medium/High | Units sold  |
| demand_level | VARCHAR   |     |                 | Demand      |

28. ROLE

| Col umn Name  | Data Type | Key | Constraints | Description              |
|---------------|-----------|-----|-------------|--------------------------|
| role_id       | SERIAL    | PK  |             | Unique role ID           |
| ro le_name    | VARCHAR   | UK  | NOT NULL    | Role name (Admin, Staff) |
| acces s level | INT       |     | CHECK ≥ 0   | Permission level         |

29. SHIFT

| Column Name | Data Type | Key | Constraints       | Description     |
|-------------|-----------|-----|-------------------|-----------------|
| shift_id    | SERIAL    | PK  |                   | Unique shift ID |
| shift_name  | VARCHAR   | UK  | NOT NULL          | Shift name      |
| start_time  | TIME      |     | NOT NULL          | Shift start     |
| end_time    | TIME      |     | CHECK start < end | Shift end       |

30. EMPLOYEE

| Column Name | Data Type | Key | Constraints          | Description        |
|-------------|-----------|-----|----------------------|--------------------|
| employee_id | SERIAL    | PK  |                      | Unique employee ID |
| name        | VARCHAR   |     | NOT NULL             | Employee name      |
| role_id     | INT       | FK  | NOT NULL             | References ROLE    |
| salary      | DECIMAL   |     | CHECK ≥ 0            | Employee salary    |
| contact     | VARCHAR   | UK  | UNIQUE               | Contact number     |
| shift_id    | INT       | FK  | NOT NULL             | References SHIFT   |
| hire_date   | DATE      |     | DEFAULT CURRENT_DATE | Hiring date        |

31. USER\_LOGIN

| Column Name     | Data Type | Key | Constraints          | Description          |
|-----------------|-----------|-----|----------------------|----------------------|
| user_id         | SERIAL    | PK  |                      | Unique user login ID |
| username        | VARCHAR   | UK  | NOT NULL, length ≥ 4 | Login username       |
| password_hash   | VARCHAR   |     | NOT NULL             | Encrypted password   |
| employee_id     | INT       | FK  | UNIQUE               | Linked employee      |
| last_login_time | TIMESTAMP |     |                      | Last login           |
| is_active       | BOOLEAN   |     |                      | Account status       |

TRIGGER AND AUTOMATION

| Trigger Name                 | Table               | Event          | Timing | Description                                         |
|------------------------------|---------------------|----------------|--------|-----------------------------------------------------|
| trg_after_sale_insert        | SALES_BILL_DETAILS  | INSERT         | AFTER  | Deducts sold quantity from inventory automatically. |
| trg_update_loyalty_balance   | LOYALTY_TRANSACTION | INSERT         | AFTER  | Updates loyalty points (earn/redeem) in account.    |
| trg_update_bill_total        | SALES_BILL_DETAILS  | INSERT, UPDATE | AFTER  | Recalculates total bill amount automatically.       |
| trg_low_stock_alert          | INVENTORY           | UPDATE         | AFTER  | Logs alert when stock reaches minimum level.        |
| trg_check_expiry_before_sale | SALES_BILL_DETAILS  | INSERT         | BEFORE | Prevents sale of expired products.                  |



# PROCEDURES AND FUNCTIONS

| Name                       | Type      | Parameters                       | Description                                               |
|----------------------------|-----------|----------------------------------|-----------------------------------------------------------|
| pr_create_restock_order    | Procedure | warehouse_id,<br>employee_id     | Creates purchase order for low-stock items automatically. |
| pr_redeem_loyalty_points   | Procedure | customer_id,<br>points_to_use    | Redeems loyalty points after validating balance.          |
| pr_get_fefo_pick_list      | Procedure | product_id                       | Suggests batch with earliest expiry for picking.          |
| fn_suggest_fefo_batch      | Function  | product_id                       | Returns best batch and days until expiry.                 |
| pr_apply_category_discount | Procedure | category_id,<br>discount_percent | Applies discount to all products in a category.           |
| pr_purge_expired_stock     | Procedure | employee_id                      | Removes expired stock and logs stock-out transactions.    |

# VIEWS

| ViewName                      | Description                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------|
| VW_INVENTORY_STATUS           | Displays product stock details across warehouses including batch and expiry.                                  |
| VW_LOW_STOCK_ALERTS           | Identifies low-stock items and suggests reorder quantity with supplier details. Provides detailed daily sales |
| VW_DAILY_SALES_LEDGER         | transactions including customer, product, and payment info.                                                   |
| VW_CUSTOMER_MEMBERSHIP_STATUS | Shows customer membership details along with loyalty points.                                                  |
| VW_STAFF_SALES_PERFORMANCE    | Tracks employee performance based on bills processed and revenue generated.                                   |
| VW_PRODUCT_PROFIT_MARGIN      | Calculates profit per unit and margin percentage for products.                                                |
| VW_FESTIVAL_DEMAND_PLANNER    | Displays product demand trends during festivals for planning.                                                 |
| VW_STOCK_MOVEMENT_LOG         | Maintains a log of stock movements for auditing purposes.                                                     |

## RELATIONSHIP, CARDINALITY AND CASCADING

| Parent Table | Cardinality | Child Table  | ON DELETE | Meaning                                                      |
|--------------|-------------|--------------|-----------|--------------------------------------------------------------|
| ROLE         | 1 → M       | EMPLOYEE     | NO ACTION | One role can be assigned to many employees                   |
| SHIFT        | 1 → M       | EMPLOYEE     | NO ACTION | One shift can be assigned to many employees                  |
| EMPLOYEE     | 1 → 1       | USER_LOGIN   | CASCADE   | One employee has one login; deleting employee removes login  |
| CATEGORY     | 1 → M       | SUB_CATEGORY | CASCADE   | One category has many subcategories                          |
| SUB_CATEGORY | 1 → M       | PRODUCT      | NO ACTION | One subcategory contains many products                       |
| BRAND        | 1 → M       | PRODUCT      | SET NULL  | One brand has many products; removing brand clears reference |
| EMPLOYEE     | 1 → M       | WAREHOUSE    | SET NULL  | One employee can manage many warehouses                      |
| PRODUCT      | 1 → M       | INVENTORY    | NO ACTION | One product can exist in multiple inventory records          |



|                 |       |                     |           |                                                 |
|-----------------|-------|---------------------|-----------|-------------------------------------------------|
| WAREHOUSE       | 1 → M | INVENTORY           | NO ACTION | One warehouse stores multiple inventory items   |
| INVENTORY       | 1 → M | STOCK_TRANSACTION   | NO ACTION | One inventory record can have many transactions |
| EMPLOYEE        | 1 → M | STOCK_TRANSACTION   | NO ACTION | One employee handles many transactions          |
| SUPPLIER        | 1 → M | STOCK_TRANSACTION   | SET NULL  | Supplier reference removed if deleted           |
| PRODUCT         | 1 → M | REORDER_LEVEL       | CASCADE   | Reorder rules deleted if product removed        |
| WAREHOUSE       | 1 → M | REORDER_LEVEL       | CASCADE   | Reorder rules deleted if warehouse removed      |
| EMPLOYEE        | 1 → M | SALES_BILL          | NO ACTION | One employee processes many bills               |
| CUSTOMER        | 1 → M | SALES_BILL          | SET NULL  | Customer can have many bills                    |
| SALES_BILL      | 1 → M | SALES_BILL_DETAILS  | CASCAD    | Bill details removed if bill deleted            |
| PRODUCT         | 1 → M | SALES_BILL_DETAILS  | NO ACTION | Product appears in many sales                   |
| SALES_BILL      | 1 → M | SALES_RETURN        | CASCADE   | Returns deleted if bill removed                 |
| PRODUCT         | 1 → M | SALES_RETURN        | NO ACTION | Product can be returned multiple times          |
| EMPLOYEE        | 1 → M | SALES_RETURN        | SET NULL  | Removes employee reference if deleted           |
| SALES_BILL      | 1 → M | PAYMENT             | CASCADE   | Payments deleted if bill removed                |
| PAYMENT         | 1 → 1 | RECEIPT             | CASCADE   | One payment generates one receipt               |
| SALES_BILL      | 1 → 1 | RECEIPT             | CASCADE   | One bill has one receipt                        |
| CUSTOMER        | 1 → 1 | LOYALTY_ACCOUNT     | CASCADE   | Each customer has one loyalty account           |
| LOYALTY_ACCOUNT | 1 → M | LOYALTY_TRANSACTION | CASCADE   | One account has many transactions               |
| SALES_BILL      | 1 → M | LOYALTY_TRANSACTION | SET NULL  | Transaction may link to bill                    |
| MEMBERSHIP_TYPE | 1 → M | MEMBERSHIP_BENEFITS | CASCADE   | Benefits removed if type deleted                |

|                 |       |                        |           |                                                     |
|-----------------|-------|------------------------|-----------|-----------------------------------------------------|
| CUSTOMER        | 1 → M | MEMBERSHIP             | CASCADE   | Customer memberships deleted if customer removed    |
| MEMBERSHIP_TYPE | 1 → M | MEMBERSHIP             | NO ACTION | Membership depends on type                          |
| PRODUCT         | 1 → M | PRODUCT_PROMOTION      | CASCADE   | Product linked to promotions through junction table |
| PROMOTION       | 1 → M | PRODUCT_PROMOTION      | CASCADE   | Promotions linked to product through junction table |
| PRODUCT         | 1 → M | PRODUCT_DEMAND         | CASCADE   | Demand data removed if product deleted              |
| FESTIVAL        | 1 → M | PRODUCT_DEMAND         | CASCADE   | Festival demand removed if festival deleted         |
| SUPPLIER        | 1 → M | PURCHASE_ORDER         | NO ACTION | Supplier supplies many orders                       |
| EMPLOYEE        | 1 → M | PURCHASE_ORDER         | NO ACTION | Employee manages purchase orders                    |
| PURCHASE_ORDER  | 1 → M | PURCHASE_ORDER_DETAILS | CASCADE   | Order details deleted if order removed              |
| PRODUCT         | 1 → M | PURCHASE_ORDER_DETAILS | NO ACTION | Product appears in many orders                      |
| PRODUCT         | 1 → M | PURCHASE_RETURN        | NO ACTION | Product can be returned                             |
| PURCHASE_ORDER  | 1 → M | PURCHASE_RETURN        | NO ACTION | Return linked to purchase order                     |
| EMPLOYEE        | 1 → M | PURCHASE_RETURN        | NO ACTION | Employee processes returns                          |



# SCHEMA STATISTICS

| Metric                          | Count |
|---------------------------------|-------|
| Total Modules                   | 5     |
| Total Tables                    | 31    |
| Total Views                     | 8     |
| Total Triggers                  | 5     |
| Total Procedures                | 5     |
| Total Functions                 | 1     |
| Total Foreign Key Relationships | 40+   |

**Thank you**